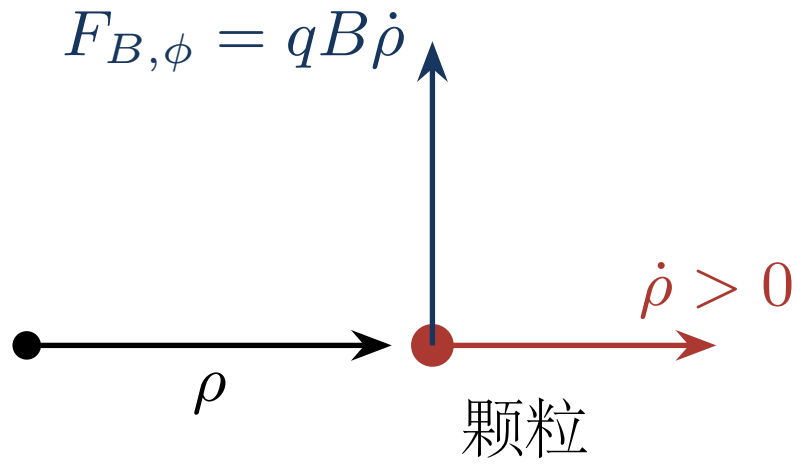

$$\dot{L} = qB\rho\dot{\rho}$$

$$J = L - qB\rho^2/2$$

机械角动量可变；磁力仍不做功